

Software documentation is not enough! Requirements for the documentation of AI

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Abstract

Purpose – Artificial intelligence (AI) is currently one of the most disruptive technologies and can be applied in many different use cases. However, applying AI in regulated environments is challenging, as it is currently not clear how to achieve and assess the fairness, accountability and transparency (FAT) of AI. Documentation is one promising governance mechanism to ensure that AI is FAT when it is applied in practice. However, due to the nature of AI, documentation standards from software engineering are not suitable to collect the required evidence. Even though FAT AI is called for by lawmakers, academics and practitioners, suitable guidelines on how to document AI are not available. This interview study aims to investigate the requirements for AI documentations.

Design/methodology/approach – A total of 16 interviews were conducted with senior employees from companies in the banking and IT industry as well as with consultants. The interviews were then analyzed using an informed-inductive coding approach.

Findings – The authors found five requirements for AI documentation, taking the specific nature of AI into account. The interviews show that documenting AI is not a purely technical task, but also requires engineers to present information on how the AI is understandably integrated into the business process.

Originality/value – This paper benefits from the unique insights of senior employees into the documentation of AI.

Keywords Documentation, Artificial intelligence, Adoption of AI, Documentation requirements, Governance of AI

Paper type Research paper

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1. Introduction

In recent years, digital transformation has had – and continues to have – an enormous effect on consumers, companies and society as a whole (Vial, 2019). Particularly the use of Artificial Intelligence (AI) has enabled companies to both improve existing services and provide new services to customers (Alfaro *et al.*, 2019; Butaru *et al.*, 2016; Khandani *et al.*, 2010). Even though using AI can have many benefits, using AI can also have undesirable side effects. Allegations of biases against women in the AI-based credit risk model behind Apple's new credit card have led to an investigation into Goldman Sachs, the bank behind Apple's new credit card (Natarajan and Nasiripour, 2019; Webb and Martinuzzi, 2019). This example shows that any entity using AI can be held accountable for the decisions made by the AI. The case of Apple's credit card also shows that there is a need to make AI fair, accountable and transparent (FAT) (Gajane and Pechenizkiy, 2017; Vettrò *et al.*, 2019; Wachter *et al.*, 2017) and to verify that the AI complies with all applicable laws, standards and regulations.

Transparency is essential for governance mechanisms (Henriques *et al.*, 2019; Winter and Davidson, 2019). In software development, documentation guidelines are important communication artifacts and describe how software works (Clements *et al.*, 2011). To ensure that IT can be governed effectively, software documentation standards have been

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developed (Hayhoe, 2012). The documentation is also essential for the governance of the software systems because the quality of the documentation is an indicator of the maturity of the governance mechanisms (Simonsson *et al.*, 2010). However, AI is different than traditional software, which is why the documentation guidelines for standard software are less meaningful for AI documentation (Rodovold, 1999). This means that there is a need to create documentation guidelines for AI applications.

However, so far, it is unclear what requirements documentation needs to fulfill and whether existing documentation guidelines help companies meet these requirements. This paper attempts to fill this gap by defining documentation requirements for AI. The research questions are:

What purpose does the documentation of AI have? What are the requirements for documenting AI?

The paper is structured in the following way. Section 2 discusses the existing literature. Section 3 presents the methodology for the paper. Section 4 presents the results of the study. Section 5 discusses the results of the paper. Section 6 concludes the paper.

2. Background

In recent years, AI and data-driven business models changed the IS landscape. A combination of more available data, more affordable computing power, and new AI approaches has enabled AI to solve more and more complex tasks (Halevy *et al.*, 2009; Sun *et al.*, 2017). The adoption of AI started in consumer technology and marketing and is now also ready to be applied in highly regulated industries such as banking and finance (Kvamme *et al.*, 2018; Martens *et al.*, 2016). However, implementing AI in practice remains challenging. The unique characteristics of some AI models make ensuring that AI applications are FAT particularly challenging. First, AI models can take over biases that are present in the data, which may lead to unfair decisions and discrimination (Barocas and Selbst, 2016). Second, the exact human influence on the decisions made by the AI can be obscured, which can result in a lack of accountability (Diakopoulos, 2016). Third, the logic behind some AI models cannot be explained well, making the AI intransparent (Adadi and Berrada, 2018). These aspects of AI can become potentially dangerous, if AI is applied in highly regulated use cases/use cases with high degrees of criticality and are not governed well. Hence, governance mechanisms are essential for organizations to ensure the success of information systems and their alignment with business objectives and legal requirements (Dubinsky *et al.*, 2011; Ndlovu and Kyobe, 2016).

Yet, governing AI is still a challenge (Winter and Davidson, 2019) and thus an adoption barrier for AI in highly regulated industries (Königstorfer and Thalmann, 2020). Even though AI-based governance mechanisms for governing organizations and people exist (Katzenbach and Ulbricht, 2019; Shin *et al.*, 2019), the literature on governance mechanisms for AI is scarce (Winter and Davidson, 2019). In this regard, audits of IT systems are important, because they verify that the company complies with applicable laws and industry standards, such as the 1028–2008 - IEEE Standard for Software Reviews and Audits (IEEE, 2008). However, even though research on applications of AI for audits already exists (Dagilienė and Klovienė, 2019; Gal *et al.*, 2016; Rezaee and Wang, 2019), research on AI audits does not focus on ensuring legal compliance which is essential for many governance approaches. For instance, SMACTR, an internal audit framework proposed by Raji *et al.* (2020) focuses on ensuring fairness and accountability in the context of AI but does not consider legal requirements. As documentations are an important basis for IT audits, they seem a very promising starting point for governing AI.

Even though the documentation can only be one component of an effective AI governance mechanism, the documentation is a promising starting point for the governance of AI. Gebru *et al.* (2018) also point out that a good documentation of the training data can aid the

communication between and also increases transparency, accountability and reproducibility. [Hutchinson et al. \(2021\)](#) argue that an effective and complete documentation of the data collection and the data maintenance system is essential for making AI accountable in both, the short- and the long run. [Diakopoulos \(2016\)](#) also argues that keeping track who was involved in the development of the AI is essential for keeping developers and other decision-makers accountable. [Cobbe et al. \(2021\)](#) point out that ensuring accountability requires information on non-technical, organizational aspects surrounding the AI and argue for the inclusion of, among other things, operational aspects of the AI systems and methods for investigating the inner working of the AI in the documentation. [Raji et al. \(2020\)](#) argue that a documentation is essential for auditing AI and ensuring compliance with FAT AI principles. However, governance mechanisms for AI in general and documentation guidelines for AI, in particular, are challenging to design, due to the significant differences between traditional software and AI. What makes AI so powerful is that AI derives models from the structure of the data, without anyone needing to tell the program what to look for or how to act ([Castelvecchi, 2016](#); [Huang et al., 2004](#)). Whereas traditional software development is characterized by programmers telling the program exactly what to do, AI applications are constructed by presenting a training program that automatically learns from a given data set ([Huang et al., 2004](#)). Because there is no direct control of the training process it is not clear how the final model behaves ([Rodovold, 1999](#)). As a result, some AI models, particularly from AI's powerful subfield Machine Learning (ML), cannot be explained easily and are referred to as "Black Box Models" in the literature and practice ([Adadi and Berrada, 2018](#); [Förster et al., 2020](#)). This black box characteristic of some AI models is a barrier to implementing AI in practice because explainability is a legal requirement for using AI in practice ([Kvamme et al., 2018](#)). Additionally, it is still unclear how suitable audit evidence related to the collection and the analysis of data in a big data environment needs to be collected ([Appelbaum et al., 2017](#)).

In the area of software engineering, documentation is the *creation and keeping current of an artifact holding information about architectural decisions for stakeholders in an accessible, nonambiguous form* ([Clements et al., 2011](#)). Even though there are documentation standards for conventional software ([Hayhoe, 2012](#)), following these standards to document AI will not result in the stakeholders having all relevant information about the AI, due to the stark difference between AI and conventional software ([Huang et al., 2004](#); [Rodovold, 1999](#)).

Even though first suggestions on how AI can be documented have been made ([Liu et al., 2020](#); [Mitchell et al., 2019](#); [Rivera et al., 2020](#); [Rodovold, 1999](#)), the authors appear to focus more on what can be documented from a technical perspective than on what needs to be documented or focus on highly specialized use cases. [Rodovold \(1999\)](#), for instance, presents a method for documenting Artificial Neural Networks ([Rodovold, 1999](#)), but does not mention how this method can help with ensuring that AI is FAT and that legal requirements are met. [Mitchell et al. \(2019\)](#) argue that their approach for documenting AI helps ensure that ethical requirements are met, but provide only a general framework that still needs to be adapted to more specific requirements from practice ([Mitchell et al., 2019](#)). [Rivera et al. \(2020\)](#) and [Liu et al. \(2020\)](#) present documentation guidelines that focus on clinical trials. Even though these guidelines are usable for the medical industry, it is unclear whether these suggestions are transferable to other use cases. Neither study describes that the exact requirements from practice for AI documentation are. Determining the requirements for AI documentation is important because research shows that the lack of documentation guidelines is not only an unsolved challenge in some application cases of AI but also a barrier to using AI in practice in highly regulated fields ([Königstorfer and Thalmann, 2020](#); [Kvamme et al., 2018](#)).

3. Method and procedure

The goal of this interview study is to determine the goals and the requirements for AI documentations. The research questions are:

RQ1. What purpose does the documentation of AI have? What are the requirements for documenting AI?

Due to the exploratory character of the study, semi-structured interviews were conducted. Due to the open-ended nature of the questions and the interview guide, the interviewees had the option to respond with non-regulatory requirements for AI documentation, such as ethical requirements. The interview guideline was tested in a pretest interview and refined even further based on the feedback.

Three groups of professionals were targeted for this interview study. First, bank professionals that were responsible for IT (multiple levels) were targeted. Their insights into the application of AI in a highly regulated industry make them suitable interview partners as the literature indicates that missing documentation guidelines are an adoption barrier for AI in regulated areas such as banking ([Königstorfer and Thalmann, 2020](#)). Second, AI engineers from software companies specialized in AI were contacted. This group of people was selected due to their knowledge of the technical aspects of AI and AI documentation. Third, consultants and regulators in the banking industry were chosen for this study. This group of people was targeted to ensure that the perspective of consultants and regulators is included as well. Two rounds of interviews were conducted. In the first round, primary insights into the topic of the study were collected. The second round of interviews was aimed at validating the findings of the first round.

3.1 First round of interviews

Between March and June 2020, 16 interviews were conducted via Skype for Business. A total of 12 of the interviews were conducted in German, the remaining 4 were conducted in English. 4 of the interviews were held with interview partners in the software industry, 7 of the interview partners worked in the banking and finance industry and 5 interview partners worked for consultancies and regulators. A total of 11 of the interview partners had more than 10 years of experience in a relevant role, while the remaining 5 interviewees had between 3 and 10 years of experience in a relevant role. The interviews were recorded, transcribed, cleansed and coded. [Table 1](#) presents some basic information about the experts that were interviewed. All but two interview partners worked in senior positions, such as CEO, CTO or Head of AI. Most interview partners had also been exposed to AI in their workplace as well as during their education. This means that they not only knew about the technological side of AI but could also see how AI affects the business side of their company.

The interviews were coded using an informed inductive approach ([Mayring, 2021](#)) taking insights from a previously obtained literature review (masked for review) into account. For instance, the structured literature review revealed that the developers of AI applications need to document how and where the AI is used in the business process of the company (masked for review). For this reason, the need to document the “Integration of the AI into the business process” became an initial code. During the interviews, new codes emerged. For example, the need to document how the strategy of the company affected the AI and the need to document the IT environment became codes as well. After the coding, the codes “Strategy affecting AI”, “IT documentation” and “Integration of the AI into the business process” got categorized as “Description of the application domain”. After the coding, the quotes used in this paper were translated and the translations were checked by both authors.

Table 1 Interview partners

<i>Expert ID</i>	<i>Industry</i>	<i>Country of employment</i>	<i>Professional experience</i>	<i>Language in which the interview was conducted</i>
1	Software	Austria	> 10 years	German
2	Banking and finance	Austria	> 10 years	German
3	Banking and finance	Germany	> 10 years	German
4	Software	Austria	> 10 years	German
5	Software	Austria	3–10 years	German
6	Software	Austria	> 10 years	German
7	Consultancy	Finland	> 10 years	English
8	Consultancy	Austria	3–10 years	German
9	Banking and finance	Germany	> 10 years	German
10	Banking and finance/Regulator	Germany	> 10 years	German
11	Consultancy	Germany	3–10 years	German
12	Banking and finance	Netherlands	> 10 years	English
13	Banking and finance	Switzerland	> 10 years	German
14	Consultancy	Netherlands	> 10 years	English
15	Banking and finance	Austria	> 10 years	German
16	Consultancy	Austria	3–10 years	German

3.2 Second round of interviews

In the second round of interviews, the findings of the first round were presented to four interviewees between April and June 2021. Two interviews were conducted in German, while the other two were conducted in English.

The results from the first round of interviews were presented in the form of a PowerPoint presentation. The interviewees were asked to comment and to give feedback on the results of the first round of interviews. The set of codes from the first round of interviews was used as a starting point for the coding of the interviews from the second round. Based on the inputs from the second round of interviews, a handful of changes were made. For instance, during the presentation of the second requirement, the documentation of the training data, the interviewees pointed out that in addition to protecting personal data, it was also essential to protect non-personal, business-critical data (i.e. data on the usage of machines, which is often used for predictive maintenance).

4. Results

The interview partners described both goals and requirements for AI documentation.

4.1 Goals of artificial intelligence documentations

The interviewees mentioned several goals of AI documentations. First, the documentation should ensure that the AI is reproducible and auditable. “A good documentation must ensure that afterward every aspect of the model can be questioned critically and analyzed. If a customer complains or a supervisory authority asks or even for an internal review we should be able to evaluate everything again and everything necessary needs to be available” (Expert 9). As the quote shows, reproducibility and auditability are important not only for internal purposes but also for legal compliance.

Second, the interviewees mentioned the goal of a documentation as knowledge artifact. The documentation should support new employees in learning about the AI and in starting to work with the AI. “A good AI documentation is a documentation that I can hand to someone and that person can start working with the AI easily. As I said, in case I lose an employee who developed this, or he is no longer available to explain the model, then someone else has to be able to use this documentation and the code to figure the AI out quickly and

to be able to work with it efficiently.” (Expert 9). The last quote shows the primary intention for the interviewee is to keep the knowledge about an AI system independent from humans in form of an explicit knowledge artifact.

Third, the interviewees state that the documentation should also be a communication artifact for external parties. One example mentioned was the documentation as a tool for communicating with investors and funding agencies. “Regarding the documentation for the research funding: There is a final report where we need to describe: What was the aim of the original funding? How far have we come? Where did we encounter problems? Where did we encounter challenges? What solutions did we find? It’s kind of like a report that you write at the end” (Expert 5). The quote shows that our interviewees did not only think of a technical documentation, but also an expert interpretation showing the strengths and limitations of the AI.

Fourth, the documentation should be an approach to ensure that the AI behaves ethically. “Our documentation requirements are based on the values defined in our Ethics Guide. We have defined principles as to how we believe AI models should work and the documentation is also based on these principles. For us, the documentation is a tool to implement the principles in our Ethics Guide in the best and most credible way” (Expert 4). Thus, the documentation should not purely document system characteristics, but also argue how core ethical requirements are addressed.

Fifth, the documentation has the goal to support risk management and to ensure that the AI can be used safely. “Documentation. Well, in the end, it’s about being able to understand the structure of an IT system. That means creating as much transparency as possible, [...] it is always about risk management. That means it is by no means about preventing any risks. But I have to at least know all the risks, and I can only achieve that by creating absolute transparency” (Expert 10). As the interviewee highlights, transparency of the AI behavior is the core challenge. In our interviews it became clear that this transparency is frequently interpreted as “we want to understand how the system works exactly” from non-AI-specialists. This is seen as one of the core challenges for AI documentation, to document the AI in such a form that auditors or risk managers can make use of them.

Based on the formulated goals of AI documentation, we identified a set of requirements regarding AI documentation.

4.2 Requirements for artificial intelligence documentation

The structured content analysis has revealed five requirements for AI documentations. First, the documentation needs to inform the reader about the business and the IT environment in which the AI operates. Second, the documentation needs to contain a description of the data used for training the AI, as well as a description of the data preparation process. Third, documentation needs to include information on the specification with which the AI model was trained and on how the AI is re-trained with new data. Fourth, the documentation needs to be understandable for knowledgeable third parties. Fifth, the effort invested into the documentation needs to be appropriate for the situation and the documentation needs to be automatic.

4.2.1 Description of the application domain. One key requirement is the documentation of the business and the IT environment in which the AI operates. Our interviewees highlighted that the documentation of the AI should consider the entire data science life cycle which was conducted. It became clear, that the documentation of AI is not only a purely technical task but also involves the documentation of the business understanding forming the basis of the AI. One interviewee argued “So a question alone is not enough. What do you want to use the answer for? [...] What is your action? And is that a required action? That needs to be documented more than how you come to the decision or data” (Expert 12). The interviewees particularly emphasize the need to document the question that the AI is

designed to answer as well as the way that the AI is integrated into the business process. This quote also shows that the documentation of AI needs to consider the business and application context which also requires people who know the business process that the AI is involved in.

Documenting the business process that the AI is integrated into also requires keeping track of the safety process of the AI. So clear measures and evaluation criteria should be established and documented during the design phase as one interviewee pointed out: “It is not only about training a machine learning model. No. It’s basically... You need really to care of all of this and how to measure when the AI makes a mistake. Or before actually, the AI makes a mistake.” (Expert 16). Such metrics and indicators are essential to ensure that the AI behaves desirably. Building on this, it is important to document how AI is embedded in organizational processes and responsibilities. This is particularly important as automated decision-making without human involvement could violate regulations such as §22 of the GDPR. One expert explains the relationship for the documentation of AI: “And from my point of view, what is even more important is the decision that I make based on the AI-based predictions. The decision I make based on the AI-based prediction, is there another person who verifies or falsifies this decision? Is there a monitoring process for this entire AI?” (Expert 11). The embedding and interaction with human deciders are highly relevant for determining potential risks and compliance. This is particularly relevant in situations in which the AI application makes decisions that could have significant ethical implications for people.

However, too much documentation could also have unintended consequences. The risk of losing the competitive advantage if the documentation of the business logic and the purpose of the AI system is described in detail was mentioned. One interviewee highlights this: “It is exactly our secret. If we were to disclose our entire model, then everyone could copy it. Of course, you don’t want that. [...] Documentation in the form of a 50-page document showing exactly how the algorithm works will never exist in portfolio management. Because we don’t want customers or other people to know how it works” (Expert 13). This quote shows that the documentation must also consider the need to protect business-critical knowledge.

4.2.2 Description of the training data set. As the AI is established on a training data set, it is necessary to document the data used to train the AI. In our analysis, it became clear, that documenting the training data set is a major difference from traditional software. As a training data set is prepared upfront, the data-selection and preparation process are also highly relevant for the documentation. Such a decision might be a reason for biases that may impact the decision-making process of the AI. One interviewee mentioned in this regard: “Basically, AI applications should be documented in such a way that their results can be reproduced. In this regard, one must, among other things, record which training data was used for the AI model. Regarding the training data, one should also record whether there were imbalances in the training data with regards to the target categories or how these imbalances were corrected” (Expert 4). The expert makes clear, that the training data set is a result of a preparation process and that it is not sufficient to just document the result of this process.

In this context, the interviewees also point to the tradeoff between making the training data transparent and the protection of personal and business-critical data. “For the end-user, it would be interesting to see which data an AI model was trained with. However, since the data is often personal information, it is not possible to make this information available to everyone. Therefore, it is necessary to make this data transparent without revealing the actual data” (Expert 4). Thus, documentation guideless should not only document the data set itself and the preparation process, but they also need to take care of privacy concerns and the protection of intellectual property and must find a suitable trade-off.

4.2.3 Descriptions of design decisions. In addition to the description of the data and the application domain, descriptions of technical design decisions made during the development of the AI are essential. These technical design decisions, such as parameters or ML algorithms are essential for the final AI model. Similar to the data set, the reasoning for selection needs to be documented as this is essential for auditing the AI. One interviewee points out in this regard: “Basically, AI applications should be documented in such a way that their results can be reproduced. [...] In addition to the information regarding the data, technical decisions such as the architecture of the model as well as the algorithms and hyperparameter settings used should be documented in a comprehensible manner” (Expert 4). What the expert highlights with “comprehensible manner” is to describe design decisions and the reasoning and not just the simple parameter settings.

However, our interviewees mentioned the fact that most AI models are not static and instead continuously developing with new data. But it is not clear yet, how to document such a continuous model evolution process in which an AI model is re-trained frequently. Expert 12 raises a handful of challenging questions about how to deal with a continuous AI model evolution. “We work sometimes with continuously learning algorithms, that learn, for instance, the risk profile of customer every week. So that algorithm changes every week. [...] So that is some dynamism [...] if I do those training aspects every week or every day or every month, do I have to account for every time I learn a new model? [...] So my models are changing all the time. What do I document? Do I document the process or the outcome? And if the outcome is ever-changing, do I need to document a changing outcome?” (Expert 12). This situation is particularly challenging because this evolution is a result of new, additional data; different weighting of new data; and possibly changing hyperparameters. For documenting continuously changing AI models, methods for dealing with this dynamism need to be found.

4.2.4 Understandable documentation for knowledgeable third parties. The interviews also revealed that the documentation still needs to be understandable for a knowledgeable reader from outside the development team. Expert 8, a consultant for financial institutions, points out that this can be a challenge. “You have to justify the models that you have used [...]. The complexity that lies in these systems can hardly be described or only with great difficulty and only through extensive documentation, which, however, must finally be understood by the reader. And I think that’s a bit of a problem right now, too. How can you document something like this so that a third party, a reasonably knowledgeable reader, can understand it?” (Expert 8). The interviewee points to an important aspect, namely, that a purely descriptive documentation is difficult to understand for external parties. Thus, typically technical details are linked and interpreted in the light of documentation standards. It is argued why a certain system fulfills a requirement or not and the knowledgeably third party can evaluate this claim. But, because of missing standards, creating understandable documentation is a particular challenge in the context of AI.

Attractiveness and especially readability is another aspect because a documentation needs to be interpreted, as one interviewee points out: “People are in general not interested in [...] They are not interested in technology. People are interested in other people or modern technology. [...] So it’s very difficult to get people involved. You really need to make [the documentation] very nice” (Expert 14). These quotes show that ensuring this understandability is not only a requirement but also a challenge that needs to be addressed.

An additional challenge is the unclarity of the target audience of the documentation as well as the extent to which the documentation needs to be shared with business partners. A new German law requires German companies to ensure that their suppliers comply with human rights. This means that whenever AI is used within companies abroad – including in China and the USA – German companies would need to verify that the compliance of a supplier’s AI with human rights. “At least in Germany, there is a supply chain law or supply chain due

diligence law. The companies are then jointly responsible for the sustainability of your entire supply chain or many tiers of your supply chain. This also applies to technology. That means I also need to know how to ensure that the algorithm from my pre-pre-supplier is working properly. If it might have an impact on my business model. [...] For me, the third parties are not just the employees and the auditors, but also my business partners. [...] you have to think about how you can ensure that the AI does not, for example, have any human rights violations in its selection or whatever” (Validation Interview 4). Additional clarification on how situations such as these should be handled are needed from the regulator.

4.2.5 Balancing the benefits and efforts of documenting artificial intelligence. In our analysis, it became clear that a suitable balance between the benefits of the documentation and the efforts invested into the documentation needs to be found. This trade-off is not new for documenting of AI but needs to be taken into consideration for implementation in practice as one interviewee points out: “If I start to document a model, then I de facto have to prove every small change in the input or in the output makes the model better. That means, I am tying up enormous resources in my development team just for documentation and I will hardly improve the models anymore” (Expert 15). The expert not only points to the high documentation effort but also to the challenge of how to deal with continuous model developments as small changes can cause big effects on the model outcome. Thus, especially the challenge of documenting continuously developing models seems a challenge for documentation also from the perspective of efforts.

Further, we found that inflexible documentation standards could hinder technical progress and innovation. Also, this trade-off between technical flexibility and regulatory requirements is not new, but it is not clear how this needs to be designed for AI. One interviewee explained: “And if you document it, in a too detailed level, suddenly within the bank and within some areas we work in, you come into a situation that you cannot change anything anymore. [...] Because that’s why we get stuck in a lot of processes that suffer. It couldn’t be changed because the regulator or documentation doesn’t allow us to change quickly” (Expert 12). Thus, designing documentation guidelines and AI audit procedures need to address this issue and find AI-specific requirements and regulations.

One approach would be that the documentation is supported by IS and integrated into the business process in a way that allows companies to adapt quickly. Similar approaches are used in traditional software engineering, but it is not clear yet how this could look like for AI as one interviewee mentioned: “The potential of automated documentation, what you are talking about is not just a potential, it is a requirement. If you think about it, if you suddenly have 10,000 decisions or 400,000 gearboxes to measure and need to process a ticket for each gearbox, or whatever, and every run through this model is somehow documented or every day, when it is retrained, that too has to be automated. [...] Otherwise, you will not be able to handle this large amount of data and generally these runs” (Validation Interview 1).

5. Discussion

We showed that documentation of AI for governance purposes is still unsolved and challenging. In this paper, we contributed to this open issue by identifying the core goals of AI documentations and by identifying the core requirements of documenting AI. First, our interviewees state that suitable documentations are essential to ensure reproducibility and auditability of AI. Second, interviewees argue that the documentation should act as a knowledge artifact for internal purposes and as a communication artifact for external stakeholders. Third, the documentation needs to be interpreted and linked to risks and ethical requirements to support risk management and the evaluation of ethical requirements. On the one hand, the goals focus on achieving FAT AI (Förster *et al.*, 2020; Jobin *et al.*, 2019; Miller *et al.*, 2017; Raji *et al.*, 2020; Smit *et al.*, 2020), and on the other hand on standard documentation goals known from literature in the area of software engineering (Bin-Abbas and Bakry, 2014; Boer and van Vliet, 2009; Forward and

[Lethbridge, 2002](#)). Thus, it can be concluded that AI documentation extends the scope of standard software documentation by focusing on the FAT principles.

In our study, we identified five requirements for the documentation of AI. The first requirement is that the documentation includes information on the environment in which the AI operates and on the actions that are being taken based on the predictions of the AI. This is in line with insights from the literature. [Shin et al. \(2019\)](#) argue that algorithms are always part of a system consisting of people and technology and that the system as a whole must be documented. [Benbya et al. \(2021\)](#) also point out that developing AI in a lab is highly different than deploying AI in real life. The lack of experience with AI deployment and particularly the lack of experience with integrating AI into business processes, is a challenge for using AI in practice ([Benbya et al., 2021](#)). [Simonsson et al. \(2010\)](#) point out that good IT governance should also provide business enabling support and that the documentation should reflect how this is achieved. At the same time, interviewees point to the challenge of protecting business-critical knowledge during the documentation of the AI and the application domain. Research needs to investigate methods and tools for balancing the benefits of documentation with the protection of business-critical knowledge.

The second requirement our study showed is that the data set and particularly the data preparation process needs to be documented. This shows in particular, that the context, provenance and limitations of the data are very important for AI systems. [Rodovold \(1999\)](#) also points to the influence of the data on the decision-making process of the AI and argues that the data needs to be documented ([Rodovold, 1999](#)). Additionally, the data preparation also has significant implications for the audit of AI. [Appelbaum et al. \(2017\)](#) point out that it is extremely challenging to collect suitable audit evidence during the audit of companies deriving value from the analysis of big datasets. So far literature mostly discusses the documentation of the data set itself, but documentation of the data preparation is missing so far. In this regard, it became clear that new approaches must deal with GDPR and other privacy requirements. Taking the trend to personalized user interaction, the storage of the training data sets is a particular challenge. At the same time, it seems extremely challenging to keep track of all user input data for individual models or recommendations, especially because the storage of this data creates problems with the GDPR. A promising avenue for future research would be to investigate methods for the GDPR compliant documentation of training data.

The third requirement is that important design decisions are documented comprehensively. [Diakopoulos \(2016\)](#) argues that information on how the AI was developed is necessary to ensure accountability in the context of AI ([Diakopoulos, 2016](#)). This is in line with older research on documenting AI ([Rodovold, 1999](#)). However, our study also identified that missing specifications about what needs to be documented when AI models are re-trained regularly are a particular challenge. Literature has shown that in the area of software engineering, there are already methods for documenting updates to traditional software ([Moreno et al., 2014](#)); however, those methods cannot cover the complexity of AI. Research needs to create clear guidelines and standards for documenting AI in this context. Defining and communicating suitable information on the IT architecture is also essential in the context of IT governance ([Simonsson et al., 2010](#)). However, continuously learning AI is also a challenge when accountability in the context of AI needs to be ensured. Our study showed that every change in AI needs to be justified and audited as well.

The fourth requirement is that the documentation is understandable for external experts. Our interviewees highlighted that it is extremely challenging for the developers themselves to understand what the AI is doing and this is of course even more challenging for externals. Research in the field of explainable AI indicates that AI developers need to focus more on creating explanations for the end-user of the AI and focus less on creating explanations that only the developers of the AI understand ([Förster et al., 2020](#); [Miller et al., 2017](#)). However, it seems relevant to make the documentation understandable for a broader audience.

Suitable modeling notations or documentation standards can help AI developers with creating understandable AI documentation. A highly relevant avenue for future research is the creation of such modeling notations or documentation standards. Additionally, the target audience of the documentation and the extent to which the documentation needs to be shared with business partners needs to be defined and evaluated by regulators.

The fifth requirement is that there needs to be a balance between the benefits of the AI documentation and the effort spent on documenting the AI using automatic documentation. Further, unclarities about how different designs of documentation guidelines affect technical innovation or flexibility became apparent. More research in this direction is required. Automation of documentation processes seems a promising approach in this regard. In software engineering, first attempts at automating the creation of release notes have already been made ([Moreno et al., 2014](#)). However, automating other parts of the documentation of AI requires documentation guidelines for AI. Hence, research needs to investigate clear guidelines on how AI should be documented.

Overall, all our interviewees and the five requirements clearly show the uncertainty when it comes to documenting AI. From a technical perspective, it is not clear what needs to be documented or which tools can support the documentation. From the legal and compliance perspective, it is not clear what needs to be documented so that legal requirements are fulfilled. Finally, from an organizational perspective, it is not clear who is responsible and how risk management or internal audit departments deal with AI.

Even though our results are largely in line with existing documentation guidelines, there are significant differences. Even though [Rodovold \(1999\)](#) also emphasizes the need to document the data, the technical design decisions made in the development of the AI and the tests of the AI, our analysis has concluded that the documentation of the integration of the AI into the business process is also a requirement for the documentation of AI. While our interviewees have stated that they are already using continuously learning AI algorithms in their organizations and argue that the topic of documenting continuously learning AI algorithms needs to be researched, the papers of [Liu et al. \(2020\)](#) and [Rivera et al. \(2020\)](#) cannot find tangible examples of continuously learning AI algorithms in healthcare and explicitly exclude recommendations on how to document continuously evolving AI models from their guidelines. This shows that existing guidelines do not fulfill all requirements from practice and that updates of these AI documentation guidelines are needed.

In addition to the requirements for the AI documentation, our study showed that being able to explain AI models is a requirement for using AI in practice. This is in line with arguments from the literature ([Khandani et al., 2010](#); [Kvamme et al., 2018](#)). However, most interviewees argue for explaining individual decisions made by the AI, instead of explaining the AI model as a whole. This is an important distinction for two reasons. First, [Adadi and Berrada \(2018\)](#) point out that the explanations of the decisions of an AI model differ, depending on the prediction that was made by the AI. Because this sort of explanation is not an architectural decision that was made in the creation of the AI, it should not be part of the documentation. Second, as [Adadi and Berrada \(2018\)](#) states that the exact methods used for explaining AI differ, depending on whether the entire AI or a single decision needs to be explained.

Finally, our study is not without limitations. First, given the exploratory character of the study, the results are not generalizable. Second, several statements from the experts had to be translated from German into English. Even though all translations have been checked by both authors, translation mistakes could still have happened. Third, the study had a strong focus on banking and finance and could have benefited from interviewing experts from companies from other highly regulated industries. Fourth, the interviewees work in Germany, Austria and The Netherlands. Even though the laws in highly regulated industries within the EU are to some extent standardized, highly regulated industries outside the EU may need to comply with different laws. An investigation of potential differences between

the documentation requirements in the EU and other countries would, therefore, be an interesting avenue for future research.

6. Conclusion

The results presented in this study offer insights into the goals of AI documentation and the AI documentation requirements of practitioners. Overall, all our interviewees and the five requirements clearly show the uncertainty when it comes to documenting AI. From a technical perspective, it is not clear what needs to be documented or which tools can support the documentation. From the legal and compliance perspective, it is not clear what needs to be documented so that legal requirements are fulfilled. Finally, from an organizational perspective, it is not clear who is responsible, how AI can be governed and how risk management deals with AI.

These insights are useful for researchers as well as practitioners. AI developers could benefit because the insights from this study provide them with a clear view of what requirements their AI documentation needs to fulfill. This can help with the creation of suitable AI documentation standards. Managers responsible for IT governance benefit because the insights from the paper allow for the creation of more targeted audit guidelines.

In the future, researchers should develop documentation guidelines that can meet the requirements specified in this paper. Additionally, research could create tools to make the documentation of AI more efficient. For instance, interviewees suggested researching methods for automatically documenting AI models.

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